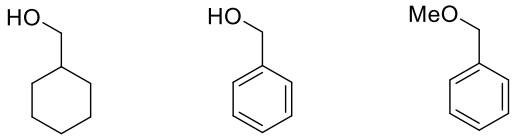
Cume for Organic Chemistry, August 2021
20 points to pass/30 points total

1) **Draw the structure of the following compounds**
   a. **m-ethylphenol (2 points)**
   b. **2-chloroanthracene (2 points)**

2) **Give the major product(s) of the following reactions:**
   a. **Benzoic acid – HNO_3/H_2SO_4 (3 points)**
   b. **Ethylbenzene + $Br_2/FeBr_3$ (3 points)**

3) **Rank the following anions in order of <u>decreasing</u> basicity (4 points):**

4) **Show how the following compounds could be synthesized from benzene:**
   a. **m-bromobenzonitrile (5 points)**
   b. **p-cresol (5 points)**

5) **How could you distinguish among the following compounds using**
   a. **Their infrared spectra (3 points)**
   b. **Their 1H NMR spectra (3 points)**